

HONEY DEFENDER

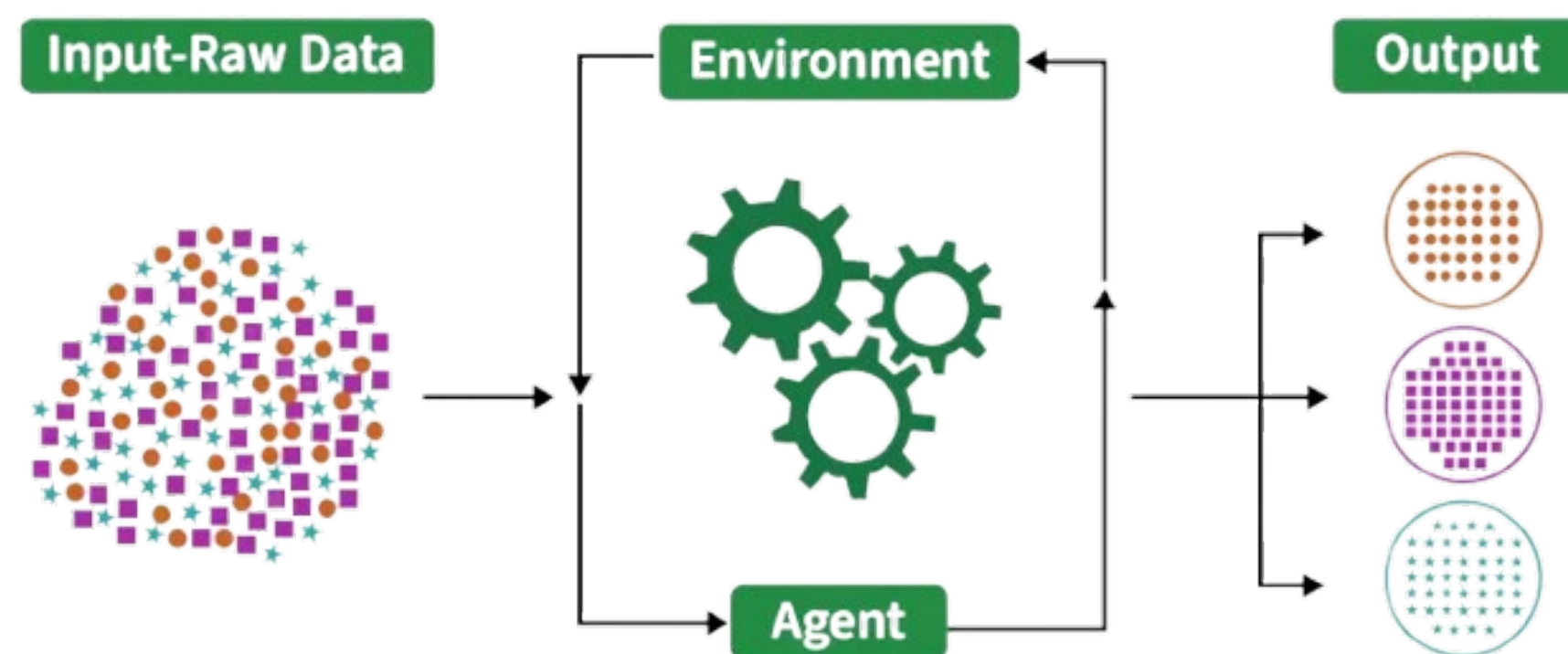
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State Based Reinforcement Learning

Reinforcement learning is a type of Machine Learning. In particular, it is the idea of an AI improving and training based on exploration and exploitation. At the beginning of its training, and based on set parameters, the AI will have a phase of completely randomized movements. This is the exploration phase; it is where the AI tries to find as many possible states as possible (since we are specifically using state-based reinforcement learning in our game). After a certain point, the AI will choose its moves on the collected data—it will choose moves correlated with wins and avoid those correlated with losses. However, this latter part is dependent on how the AI is rewarded.

Reinforcement Learning in ML

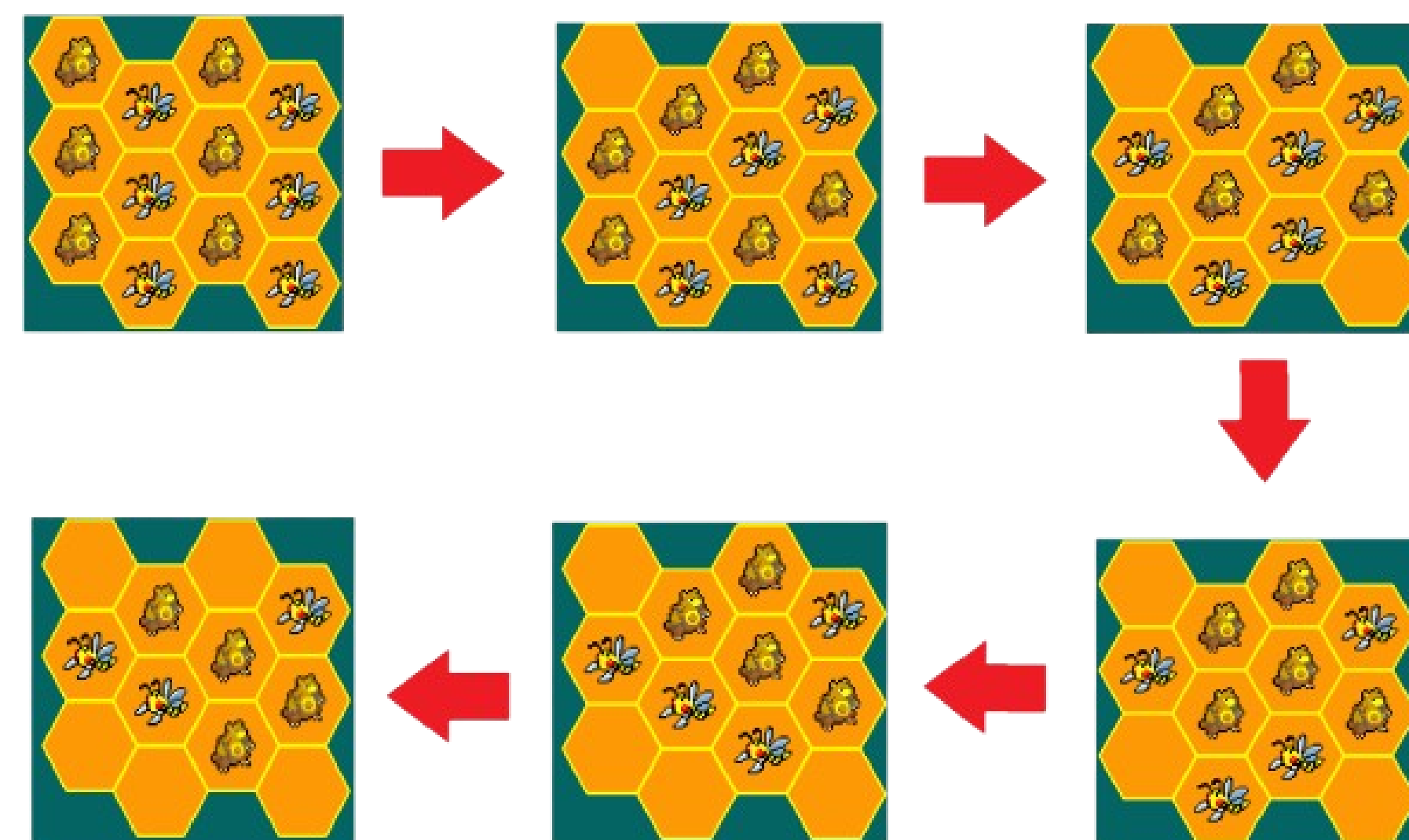


How Our AI is Trained

The AI is rewarded for wins and kills, and punished for ties and losses. If both AIs were perfect, they would tie every time. So we gave only a slight punishment for ties and a slight reward for kills. However, since the AI would never kill because of this, we upped the reward for killing. Ultimately, we want the AI to win as much as possible, but a tie would be second best, obviously. To train the AI evenly, we started punishing the AI if it was killed; this would have hopefully created an AI that was focused on surviving, but also killing. In our project, our AI learns over time through reinforcement learning. The AI will play potential games, and based on whether they win, lose, or tie those games, the AI might be prone to repeat or avoid previous moves.

Overview and How to Play

Honey Defender is an original two-player turn-based game that has been created and used for a State-Based Reinforcement Learning Project. This game was made with heavy inspiration from the popular board game Abalone, but with many key differences. One of which is the size of the board. Our board game is dynamically sized, meaning the number of rows and columns can be easily changed. The default board has 3 rows and 4 columns. One player controls all bear tokens, and the other player controls all bees. These tokens are placed on every tile at the beginning of the game, with each column being one specific player. The goal of the game is to push off all but one of the enemy tokens from the hexagonal board. On a player's turn, they can choose to move in any of 6 directions that do not move their piece off the board.



Acknowledgements & Contact Info

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